Kuiyi (Cory) Gao

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Profile

Computer science undergraduate building generative AI and agent systems across retrieval, fine-tuning, evaluation, and runtime safety. I develop Python pipelines, browser-based tasks, automated judges, and auditable guardrails, and carry work from prototyping through testing, analysis, and technical communication.
Languages: Mandarin (native), English (TOEFL iBT 110), Cantonese (intermediate).

Education

University of North Carolina at Chapel Hill Jan 2026–May 2027
B.A. in Computer Science, GPA 3.68/4.00 *Chapel Hill, NC*
Coursework. Algorithms, machine learning, natural language processing, technology ethics, C/Unix systems, and object-oriented design.

The Chinese University of Hong Kong Sep 2023–May 2026
Computer Science, Faculty of Engineering ELITE Stream *Hong Kong*
Coursework. Graduate deep learning, artificial intelligence, data structures, formal languages, computer organization, probability, statistics, and linear algebra.

Research Experience

RealGUINoise: GUI-Agent Robustness under Interface Noise [1] Sep 2025–present
Co-first author; CUHK, advised by Prof. Michael R. Lyu *Hong Kong / remote*
Engineering. Built paired web and Android tasks and browser automation for controlled overlays, dynamic content, layout shifts, and look-alike controls.
Evaluation. Implemented automated judging and paired measures for task completion, unsafe actions, and action-sequence changes.

RuntimeTickets: Runtime Enforcement for Third-Party Agent Skills [2] Jun 2026–present
First author; MBZUAI UGRIP, advised by Dr. Steve Liu *Abu Dhabi / Beijing*
Method. Designed task-scoped contracts and guardrails for agent tool use, with pre-execution checks and explicit PASS, BLOCK, and DEFER decisions.
Engineering. Built auditable decision and effect traces for testing and human review; lead method development, experiments, and manuscript writing.

Chain-of-Jailbreak: Multi-Turn Safety Attacks on Image Generation [4] Jun 2024–Feb 2025
Second author; Tencent AI Lab and CUHK *Hong Kong*
Research. Designed the multi-turn image-editing attack and prompting defense, contributed benchmark data, and ran most experiments.
Delivery. Wrote evaluation scripts and prepared figures, manuscript text, and rebuttal.

MMSafeAware: Benchmarking Multimodal Safety Awareness [3] Dec 2023–Oct 2024
Third author; Tencent AI Lab *Hong Kong / Shenzhen*
Benchmark. Constructed data and evaluation for 1,500 image-text pairs across 29 safety scenarios.
Model work. Ran vision-centric fine-tuning, analyzed missed risks and benign false alarms, and contributed to submission and rebuttal.

Text-KEN: Comparing Collections of Text May–Aug 2025
Summer research with Prof. Farzan Farnia, CUHK *Hong Kong*
Engineering. Built a Python pipeline that compares text collections using sentence embeddings and distribution-level statistics, without training a classifier.

Publications and Manuscripts

[1] Yongjiang Wu*, Junyuan Zhang*, Ada Chen*, **Kuiyi Gao***, Wenxuan Wang. (2026). *RealGUINoise: An Interactive Cross-Platform Benchmark for GUI Agent Robustness under Real-World Interface Noise*. **Co-first author** (*equal contribution). Under review. [arXiv]

[2] **Kuiyi Gao**. (2026). *RuntimeTickets: Source-Grounded Security Obligations for Agent Skills*. **First author.** Under review at ICLR 2027.

[3] Wenxuan Wang, Xiaoyuan Liu, **Kuiyi Gao**, Jen-tse Huang, Youliang Yuan, Pinjia He, Shuai Wang, Zhaopeng Tu. (2025). *Can't See the Forest for the Trees: Benchmarking Multimodal Safety Awareness for Multimodal LLMs*. ACL 2025 Main. [paper] [code]

[4] Wenxuan Wang, **Kuiyi Gao**, Youliang Yuan, Jen-tse Huang, Qiuzhi Liu, Shuai Wang, Wenxiang Jiao, Zhaopeng Tu. (2025). *Chain-of-Jailbreak Attack for Image Generation Models via Step by Step Editing*. Findings of ACL 2025. [paper] [code]

Selected Projects

MemSlot: Long-Document Retrieval and Compression for Question Answering [code] UNC, 2026
Purpose. Treats long-document QA as a retrieval and compression problem before OCR, under a limited context budget.
Work. Built a neural retriever with memory-slot attention, overlap-aware scoring, saliency-guided rendering, and pruning baselines.
Output. Training and evaluation notebooks, pruning comparisons, and saved word-saliency visualizations.

Poetry and Image Variation: Comparing Poetic Movements [code] CUHK, 2025
Purpose. Studies whether Imagist and Romantic excerpts produce different visual variation in generated images.
Work. Generated images with FLUX and Stable Diffusion and compared saved CLIP-based scores in Python.
Output. Generated-image grids and plots comparing the saved variation scores across poetry groups.

Movability Segmentation: Four-Class Scene Understanding [code] CUHK, 2025
Purpose. Maps scene elements into movability classes for action-oriented perception.
Work. Adapted DeepLabV3 for four-class segmentation and implemented data remapping, preprocessing, training, and checkpoint inference.
Output. Course report, training code, saved checkpoints, and prediction overlays on archived scenes.

Selected Distinctions

Dean's List, University of North Carolina at Chapel Hill Spring 2026
Professor Joseph Lau Scholarship of Innovation
CUHK Faculty of Engineering ELITE Stream Award
CUHK College Entrance Scholarship
CUHK Exchange Scholarship for UNC Spring 2026
Gaokao, top 2% in Beijing 2023

Technical Skills

Generative AI and retrieval. Transformers, fine-tuning, embeddings, RAG, dense and keyword retrieval, reranking, and long-context QA.
Agent engineering. Tool use and function calling, orchestration, context and state management, runtime guardrails, and human-in-the-loop review.
Evaluation and observability. Task design, automated scoring, regression tests, paired experiments, trace logging, and error analysis.
Software and delivery. Python, PyTorch, REST/JSON APIs, Git, Linux, Docker, Playwright, testing, data visualization, and technical writing.